



OPEN LETTER

A Practical Checklist for Embedding Inclusivity in Advanced Behavioural Research Labs

[version 1; peer review: 2 approved with reservations]

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Abstract

Despite growing recognition of the importance of inclusive research, behavioural studies are often conducted on narrow, unrepresentative populations. This systematic exclusion is reinforced by where and how research is conducted. Inaccessible lab environments, complex procedures, and limited accommodations can inadvertently create barriers to participation for individuals with long-term health conditions, neurodivergence, caregiving responsibilities, or heightened anxiety. We present a practical checklist to support researchers in designing more inclusive lab-based behavioural research. Co-produced with patient contributors during a three-year programme of patient and public involvement and engagement in the United Kingdom, the checklist prompts researchers to reflect on and address potential barriers across all stages of the research process – from protocol design and recruitment, through to participation and dissemination. By embedding inclusion from the outset, researchers can begin to reduce structural exclusion, broaden participation, and ultimately enhance equity and relevance in behavioural science.

Plain language summary

Research overview: Research studies often do not include a wide range of participants. This makes it harder for results to generalise to the whole population. People with ongoing health conditions, neurodivergence, and/or caregivers may not have the chance to take part in research. This can limit the quality, relevance, and impact of scientific findings.

Researchers worked with members of the public to find out what makes it harder to take part in research. Difficulties included travel, scheduling, tasks, and unfamiliar or unwelcoming environments.

Key takeaways: Based on this feedback, the researchers made a

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Any reports and responses or comments on the article can be found at the end of the article.

practical checklist. The checklist covers every stage of research, to help other researchers to plan studies that are more accessible for participants.

Limitations: The checklist is based on behavioural lab research. It may not cover all research settings or participant needs. It is a starting point for researchers and is not a complete solution.

Why it matters: More inclusive science can improve participation, data quality, and how useful the findings are. This is particularly the case for people who are the most affected by health conditions, but who are the least often included in research.

Keywords

Inclusive Research, Behavioural Research, Research Design, Participant Accessibility, Research Inclusivity Checklist, Underserved Populations

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Competing interests: No competing interests were disclosed.

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Introduction

Behavioural research — particularly in the context of injury and long-term conditions such as chronic pain — is often conducted on narrow, unrepresentative populations (e.g., “WEIRD” samples, [Henrich *et al.*, 2010](#)), limiting the generalisability and equity of scientific findings. Barriers owing to where and how research is conducted mean that many underserved groups are systematically excluded. Inaccessible lab environments, complex protocols, inflexible scheduling, and lack of sensory accommodations often create barriers for people with mobility limitations, neurodivergence, chronic pain, caregiving responsibilities, or heightened anxiety ([Health Research Authority, 2024](#); [NIHR, 2020](#)).

This checklist outlines practical, patient-informed strategies intended to make lab-based behavioural research more inclusive and accessible. The guidance has been co-produced through a three-year programme of patient and public involvement and engagement (PPIE) in the United Kingdom between 2022 and 2025. Activities included lab tours, interviews, workshops, and task demonstrations involving contributors with chronic pain, neurodivergence, acute injuries, and brain health conditions, alongside behavioural and neuroimaging researchers. Across sessions, participants highlighted a range of overlooked barriers — from uncomfortable equipment and sensory overload to difficulties understanding or preparing for research tasks. Some contributors described environments that felt alienating or anxiety-provoking; others raised concerns around fatigue, mobility, or unclear expectations. These themes illustrate that traditional research settings can inadvertently exclude those who may have the most to contribute — and benefit — from scientific progress.

Involving patients meaningfully in study design leads to improvements in the quality, accessibility, and relevance of research ([Brett *et al.*, 2014](#)). While this guidance is grounded in pain research, its considerations apply broadly across studies involving clinical, neurodivergent, or marginalised populations. The checklist is intended to support researchers in integrating equity, diversity, and inclusion into every stage of the research process.

Checklist

1. Getting started

- a. Patient-facing materials – Are they accessible and understandable?
 - i. Are all study materials and instructions presented in clear, jargon-free language? Aim for readability consistent with national literacy levels (e.g., UK average: 9–11 years).
 - ii. Can materials be made available in multiple languages, where appropriate? If not, is this limitation clearly justified (e.g., legal/consent requirements)?
 - iii. Are accessible formats offered across communication types? For example, live captioning for videos, large print, dyslexia-friendly fonts, icon-supported text, audio versions or screen-reader compatibility.

- b. Protocols – Are procedures practical for your population?
 - i. Does the study design consider potential limitations in mobility, task endurance, or fatigue?
 - ii. Is the session length realistic? Give options: one long session versus multiple shorter ones.
 - iii. Will any procedures cause pain, discomfort, or sensory overload? Are participants made aware of this beforehand? Can necessary aids and/or medications be maintained?
- c. Population – Are the inclusion and exclusion criteria justified?
 - i. Are all criteria evidence-based and ethically sound?
 - ii. Avoid overly broad exclusions like “no mental health history” unless scientifically necessary.
 - iii. Consider inclusivity for pregnant/lactating individuals or those with complex health profiles.

2. It's approved! Approaching potential participants

- a. How do we approach participants?
 - i. Have recruitment strategies been tailored to the population of interest?
 - ii. Is the use of digital platforms (e.g., social media) age- and culturally-appropriate?
 - iii. Can you engage national charities, local support groups, or GP/hospital settings?
 - iv. Can trusted intermediaries (e.g., clinicians, community leaders) introduce the study?
- b. Where do we find participants?
 - i. Account for neurodivergence, caregiving duties, and working patterns.
 - ii. Reach out via overlapping or adjacent communities if direct access is limited.
- c. What is the study doing and why are we doing it?
 - i. Are optional preparatory materials available (e.g., videos, social stories, or photos of lab spaces/team/procedures)?
 - ii. Would a pre-visit phone call or online meeting help to answer questions, reduce anxiety, or build rapport?

3. Getting to the session

- a. Scheduling – Is timing flexible?
 - i. Can participants book outside of typical working hours (e.g., for work or care responsibilities)?
 - ii. Can they pause, reschedule, or cancel at short notice (e.g., due to pain or mental health symptoms)?
- b. Transport – Have transportation needs been considered?
 - i. Ask participants about travel plans. Do they need accessible parking, taxis, or public transit support?

- ii. Can the session coincide with a clinical appointment to reduce travel burden?
- c. Directions – Are arrival instructions easy to follow to outsiders?
 - i. Are arrival instructions tailored to the transportation method? Provide transport-specific guidance, including photos or video guides.
 - ii. Ensure information is optional and not overwhelming.
- d. Accessibility – Is the physical space suitable?
 - i. Consider parking and building access, lift availability, waiting areas, and sensory needs (lighting, sound, smells).
 - ii. Can participants bring a companion, child, or therapy animal? Is there a space for anyone accompanying the participant to wait?
 - iii. Is childcare or another staff member available to support caring responsibilities during participation?
- e. Reimbursement – Are participants compensated fairly?
 - i. Clearly explain when and how payments will be issued.
 - ii. Use accessible payment options – cash where feasible – to ensure no one is left out of pocket.

4. Meeting the researcher

- a. Do researchers wear name badges?
- b. Do researchers show and/or introduce their pronouns?
- c. Is inclusive, respectful, and culturally sensitive language used?
- d. Has the researcher received equity, diversity, and inclusion (EDI) and ethical conduct training?
- e. Are researchers supported in navigating complex interpersonal or cultural interactions?

5. Inside the lab

- a. Comfort and safety
 - i. Are chairs adjustable and suitable for diverse needs?
 - ii. Avoid strong sensory stimuli such as loud noises, strong smells and bright lights that may be aversive to hypersensitive and neurodivergent participants.
 - iii. Ensure accessible and gender-neutral toilets are nearby where possible.
 - iv. Provide facilities for washing if needed (e.g., after physical tasks).
- b. Clarity and orientation
 - i. Remove or cover non-essential or unfamiliar equipment to reduce distress.

- ii. Use signs to clarify the function of visible machines or unusual devices.
- iii. Adapt verbal and non-verbal communication to the participants' preferred style and level of need.

6. Conducting the study

- a. Procedure – Is it clear, flexible, and respectful?
 - i. Are instructions clear and concise? Can they be delivered visually or verbally?
 - ii. Is physical assistance clearly consented to each time?
 - iii. Are group formats available to reduce anxiety (if appropriate)?
- b. Managing fatigue and wellbeing
 - i. Are regular breaks offered? Is there a quiet space or sensory room to rest?
 - ii. Consider effects of treatment withholding and plan accordingly.
 - iii. What truly needs to happen in the lab, and what can be done elsewhere?
- c. Respect for modesty and culture.
 - i. Provide private spaces for changing, if needed. Is there a space nearby for prayer or breastfeeding/to express milk?
 - ii. Notify participants in advance about tasks requiring physical contact or skin exposure.
 - iii. Accommodate requests for a single-sex research team where possible.

7. Finishing up

- a. Leaving the study – Can participants leave comfortably?
 - i. Will participants be fatigued or distressed? Is there a space for recovery before leaving?
 - ii. Have departure instructions been provided? Can participants leave in the same way that they arrived?
- b. Post-study support – What happens if issues arise?
 - i. Provide contact details and clear instructions on how to report any problems or follow up.

8. Getting the results back to the people

- a. Do you have permission (i.e., ethical approval) to recontact participants with results?
- b. Offer summaries in accessible formats (e.g., short video, infographic, easy-read document) and provide access to the full report (i.e., published paper).
- c. Can results be shared publicly (e.g., blogs, podcasts, local news)?
- d. Consider participant-led feedback sessions or focus groups to discuss findings.

Summary and what's next

Embedding a culture of inclusivity into research design is essential for producing diverse, equitable, and representative science — particularly for groups who have been historically excluded or underserved. This checklist offers a structured approach for identifying and reducing barriers to research participation, covering the entire research journey from applying for ethical approval through to dissemination of findings. The checklist is grounded in the experiences of people living with or caring for those with chronic pain, neurodivergence, and brain health conditions, to develop a wide range of practical considerations which may be overlooked in traditional lab-based behavioural research.

Crucially, inclusive planning should begin before a study starts through meaningful involvement of the communities it aims to serve. While some limitations (e.g., resources, legal, or safety-related concerns) are unavoidable, researchers should adapt their practices to promote accessibility wherever they can. This is not an exhaustive list, but rather a resource to prompt reflection, support action, and contribute to ongoing

conversations about how to build more inclusive behavioural research environments now and in the future.

Disclaimer

The views expressed in this article are those of the author(s). Publication in Wellcome Open Research does not imply endorsement by Wellcome.

Data and software availability

No data are associated with this article.

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[Reference Source](#)

Open Peer Review

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Cathy Beresford 

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This paper is an open letter which details a practical checklist for embedding inclusivity in advanced behavioural research labs. The authors outline the importance of inclusive research and highlight that behavioural studies are often conducted on narrow, unrepresentative populations. Those with long-term health conditions, neurodivergence, caregiving responsibility or heightened anxiety frequently face barriers to participation. The authors offer a checklist of practical, patient-informed strategies intended to make lab-based behavioural research more inclusive and accessible.

While the authors explain that the checklist was co-produced through a three-year programme of patient and public involvement and engagement (PPiE) in the United Kingdom between 2022 and 2025, the paper does not describe who took part in this process (e.g. number of contributors, relevant lived experiences, or roles in shaping the checklist). Providing this information would help readers understand whose perspectives informed the checklist and would enhance transparency.

However, the open letter is well-written, with a clear abstract and well-explained points. I particularly like the plain English summary.

In the Introduction, "WEIRD" samples are cited. It will be helpful if this is explained so that the reader understands without having to read about it elsewhere.

The checklist is detailed but clear and will be useful to researchers working in behavioural research and beyond.

Is the rationale for the Open Letter provided in sufficient detail?

Yes

Does the article adequately reference differing views and opinions?

Partly

Are all factual statements correct, and are statements and arguments made adequately supported by citations?

Yes

Is the Open Letter written in accessible language?

Yes

Where applicable, are recommendations and next steps explained clearly for others to follow?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Qualitative research, PPI, long-term conditions, inclusive research, empowerment

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Reviewer Report 21 November 2025

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Natalie L Marchant 

University College London, London, UK

The context for this checklist addresses the significant need to improve inclusive practices in research, and addresses a gap in current guidance by focusing on lab-based behavioural research. The letter highlights the importance of co-production with patients from the outset and that they co-produced this checklist which is brilliant. The team also acknowledge that this checklist is grounded in pain research and co-produced with people living with specific conditions, but that it could be applicable to other lab-based research and people living with other conditions. The letter is also written in a clear and accessible manner.

General

The checklist could be greatly improved by providing links to other resources that would support items mentioned here. This would ensure that it is used, stands out from other existing checklists, and better supports researchers to improve their inclusive practices. For example, the HRA and others provide more detailed guidance on making information sheets and consent forms more accessible; other checklists and resources provide complementary information and considerations (e.g. TrialForge resources (although trial focused) / NIHR INCLUDE resources, Miles D Witham et al.

BMJ Open 2020;10:e043634). There is also literature about determining eligibility criteria.

I think an aspect that could be included in the next steps or other considerations section relates to how researchers can reach and recruit participants. While adding this to the checklist may be beyond the scope of the letter, mention of how researchers should be encouraged to at least consider 'what's in it' for these historically underrepresented participants (why should they give up their time to participate?) would be welcome.

Specific

While I agree with the authors' statement about "WEIRD" samples, this checklist does not address that limitation of research ie addressing research conducted in western, educated, industrialised, rich or democratic settings. I would recommend using a different example and reference that speaks to the lack of diversity based on participant characteristics rather than setting.

I also agree with the sentiment of this statement "Ensure information is optional and not overwhelming" however who decides what is "overwhelming" and how would researchers know? The authors mention co-production in the introduction and summary, and it would be helpful if this was tied to some checklist items as they would inform decisions such as what is "overwhelming". Could acknowledgement of this ambiguity and the need for co-production to help inform checklist decisions be made explicit?

Regarding this statement: "Is childcare or another staff member available to support caring responsibilities during participation?" – important to note that the staff member would need to have required certifications / DBS checks.

Is the rationale for the Open Letter provided in sufficient detail?

Yes

Does the article adequately reference differing views and opinions?

Partly

Are all factual statements correct, and are statements and arguments made adequately supported by citations?

Partly

Is the Open Letter written in accessible language?

Yes

Where applicable, are recommendations and next steps explained clearly for others to follow?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: My area of research involves improving inclusive research practices particularly regarding ethnically-minoritised groups in ageing and dementia research.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.
